

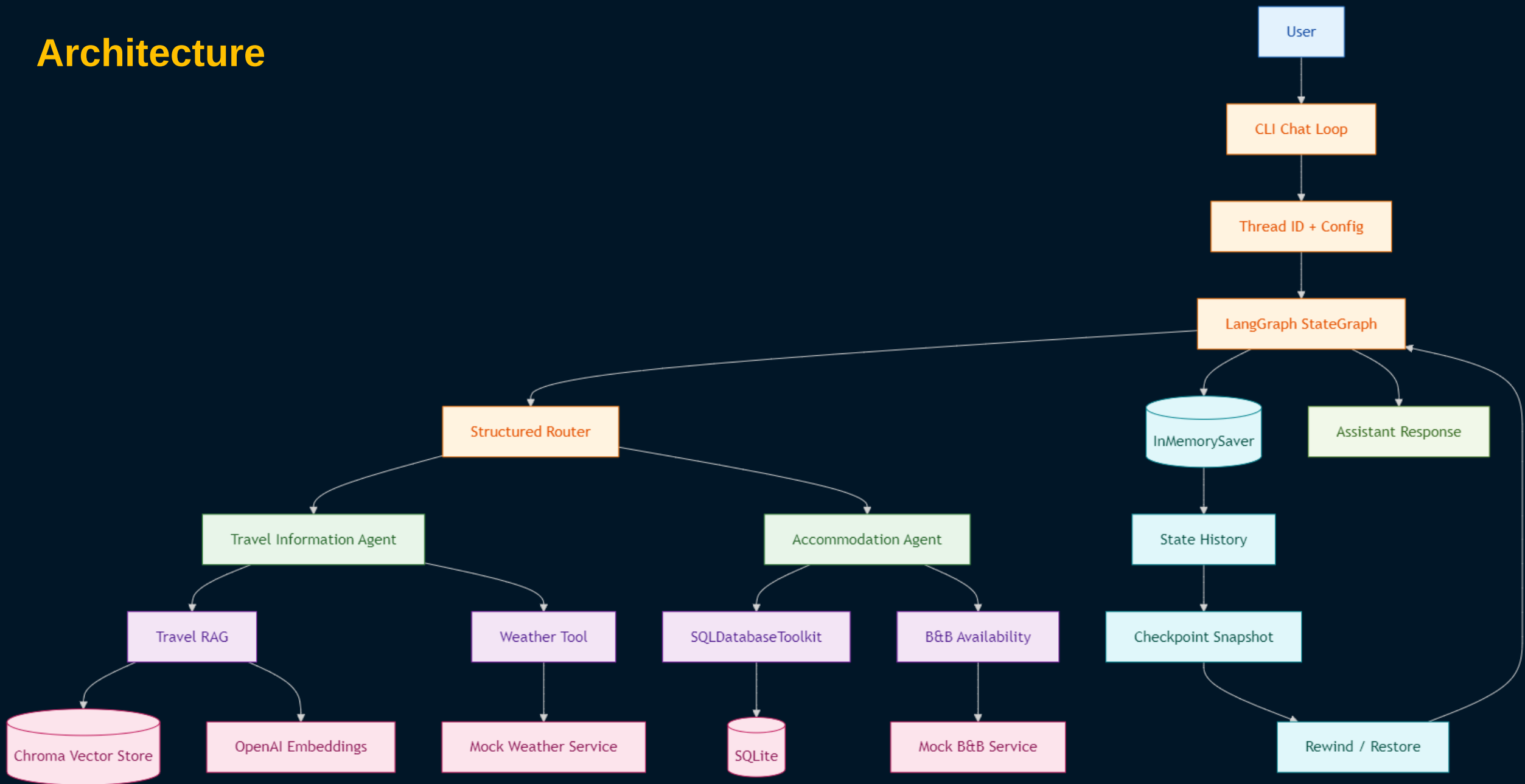
Agentic AI Engineering

***Stateful Multi-Agent AI Platform
with
LangGraph Checkpoint Inspection & Rewind***

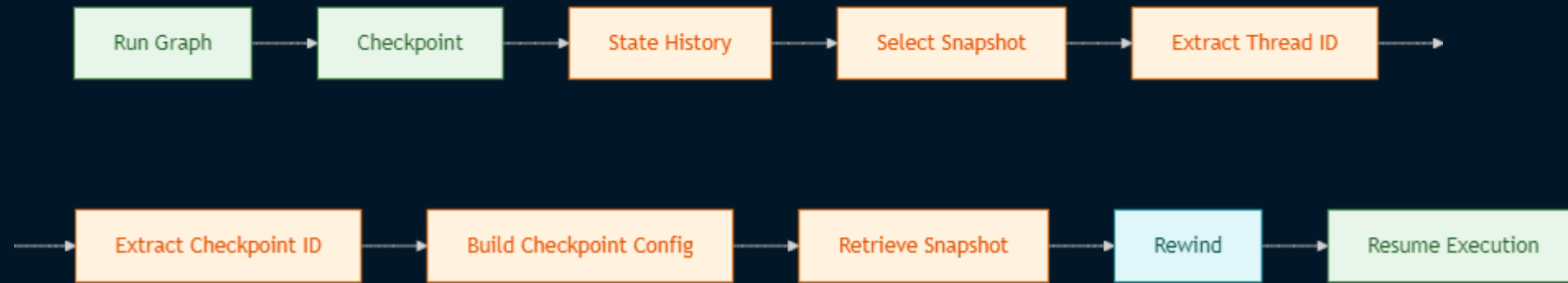
Max Montaseri

- Architected a **stateful multi-agent Agentic AI workflow** using LangGraph StateGraph, typed state, Command routing, thread management, and checkpoint-based short-term conversational memory.
- Implemented **checkpoint state inspection and rewind capabilities** using LangGraph `get_state_history()` and `get_state()`, enabling developers to inspect historical execution state and resume workflows from selected checkpoints.
- Designed a checkpoint-aware debugging workflow that extracts `thread_id` and `checkpoint_id`, reconstructs checkpoint configuration, restores historical state, and executes follow-up requests against the restored context.
- Engineered structured LLM routing with **Pydantic models and Enum-constrained destinations**, converting model classification into deterministic graph transitions between specialized travel and accommodation agents.
- Integrated **RAG/Chroma, OpenAI embeddings, SQLiteDatabaseToolkit, SQLite** and **custom tool interfaces** into domain-specific specialist agents.
- Integrated the **OpenAI Responses API with previous-response IDs** alongside LangGraph checkpoint state to support model-level conversational continuity and graph-level state management.
- Established a foundation for **state-aware Agentic AI debugging, replay, regression testing** and **workflow recovery**, while identifying durable persistence, security, and observability requirements for production deployment.

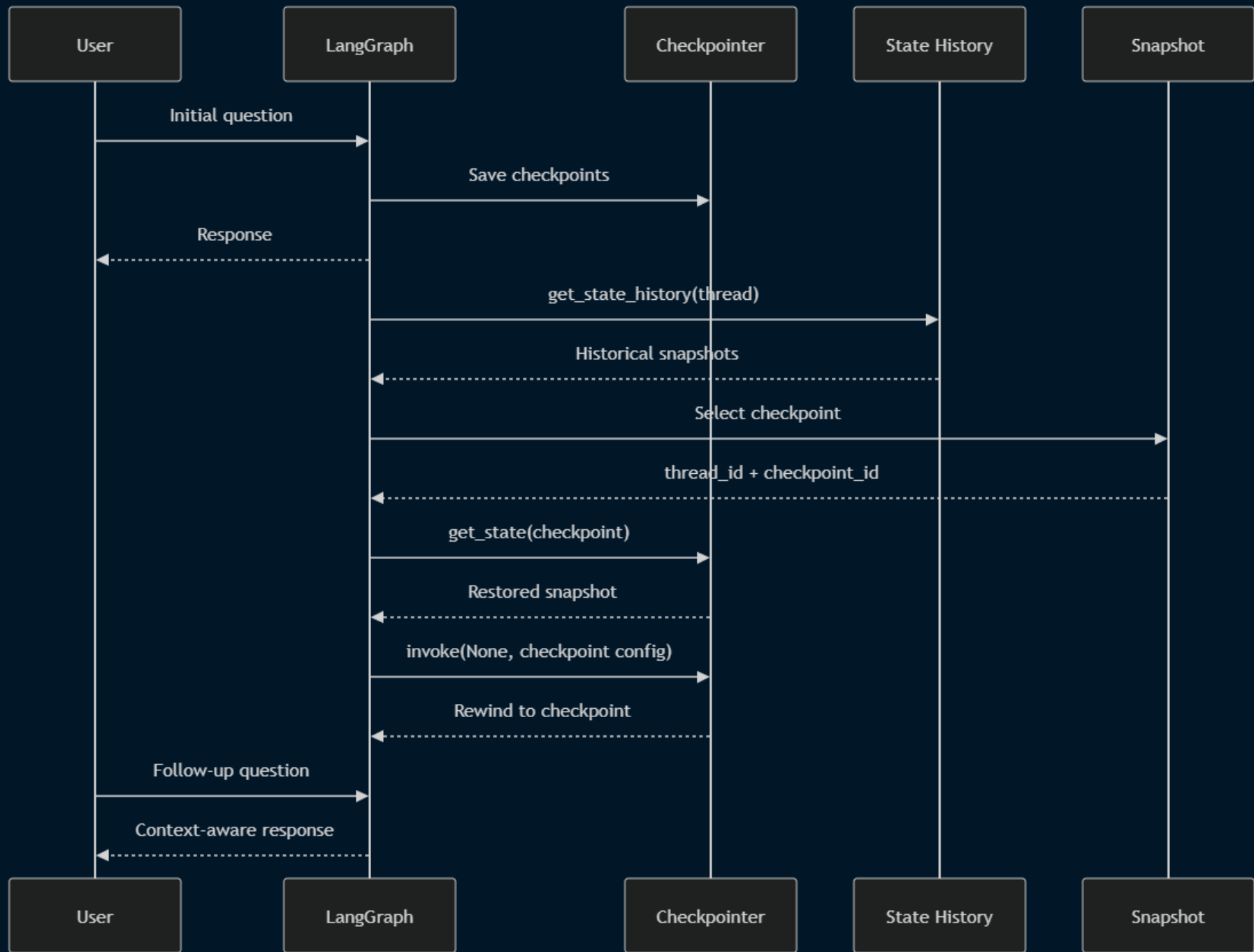
Architecture



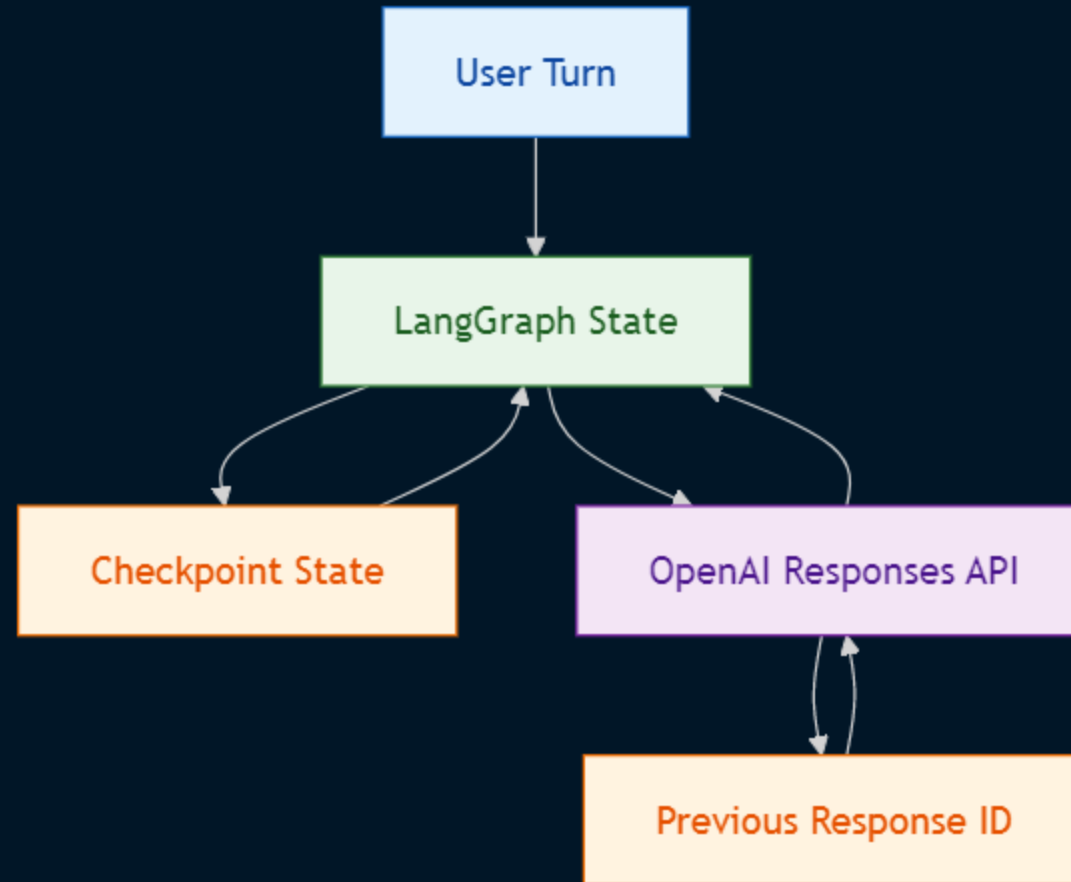
State Inspection Workflow



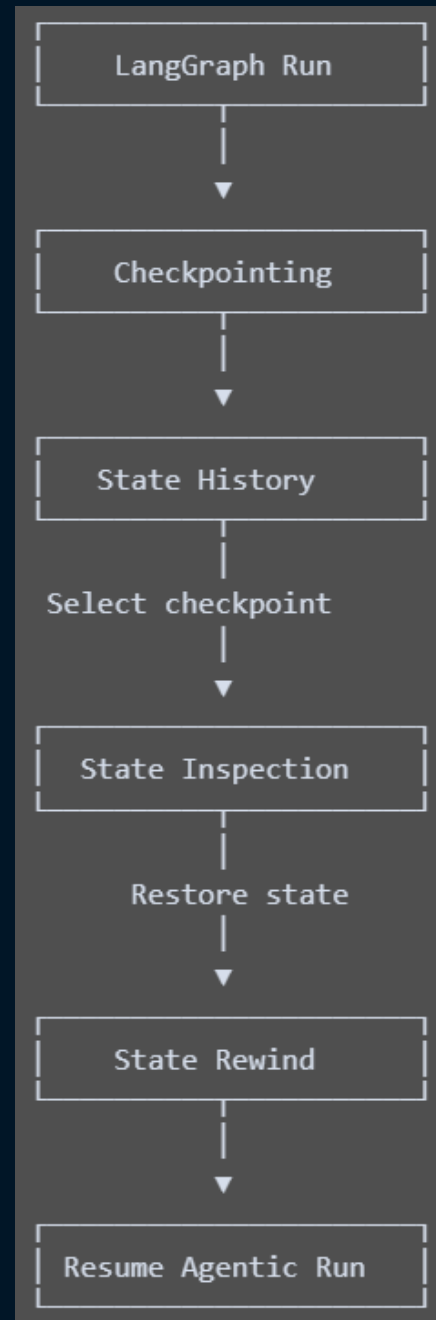
State Inspection Sequence



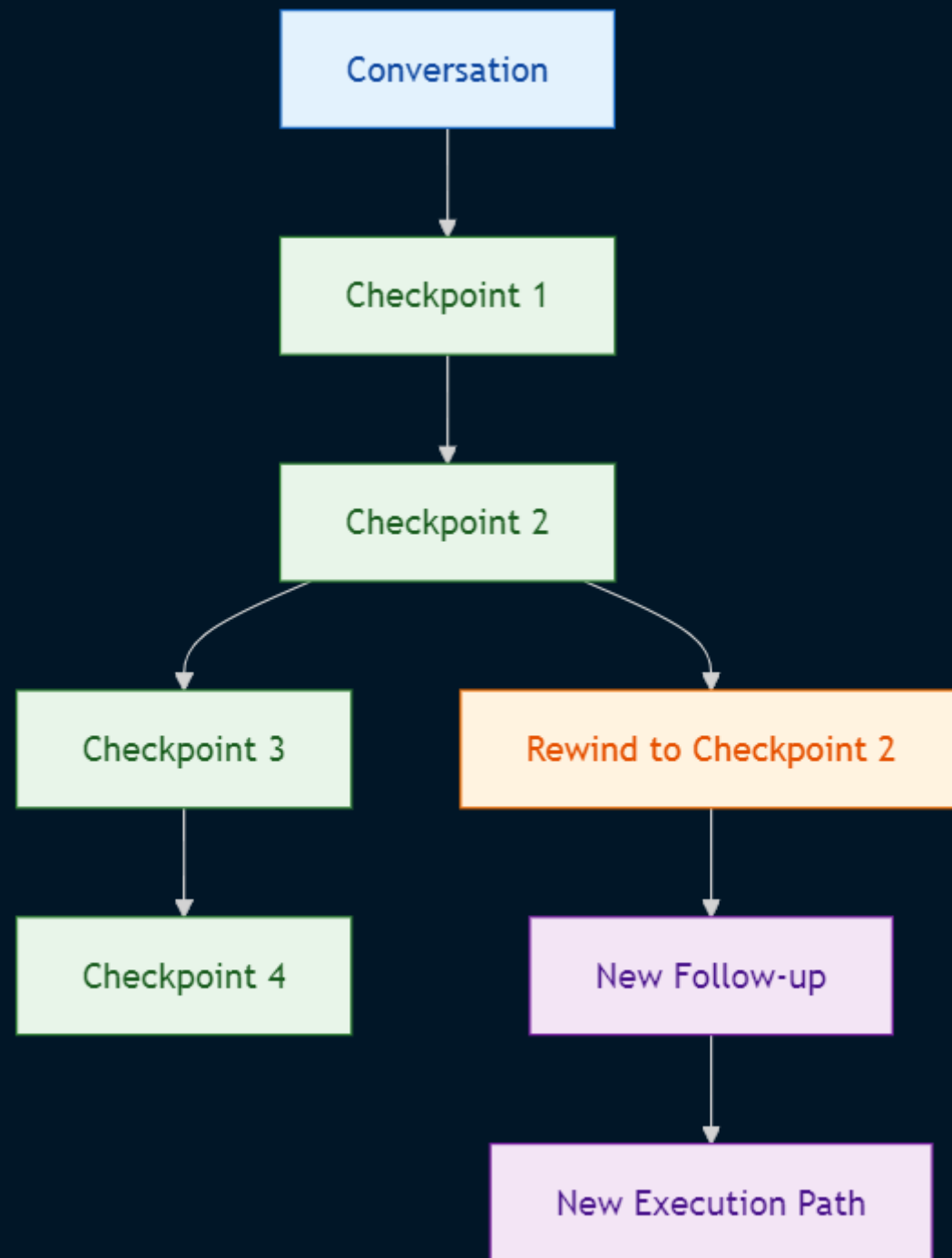
Two-Layer State Model



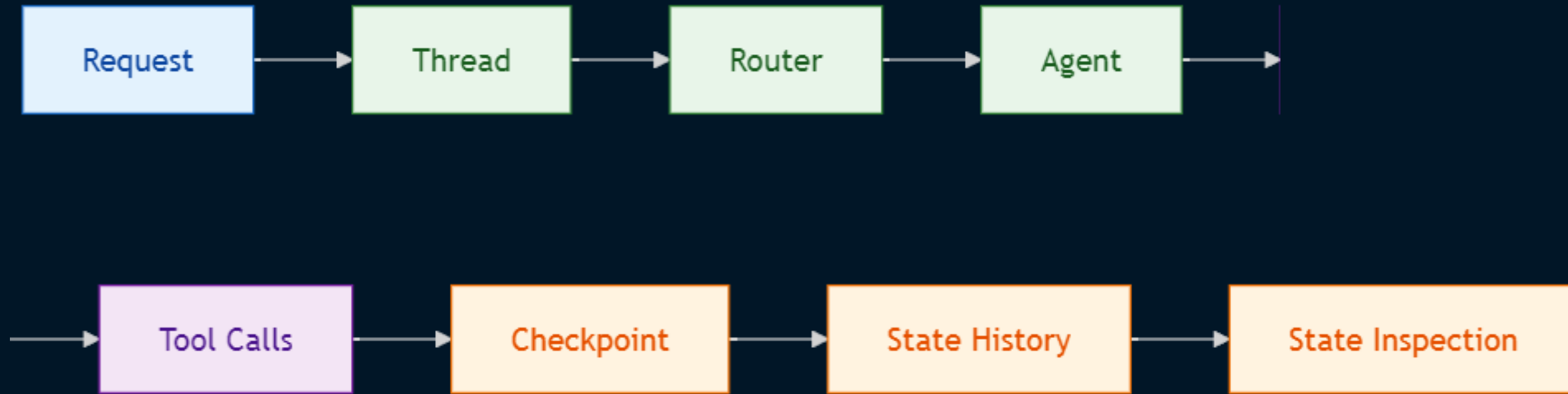
State Inspection Architecture



Rewind as a Development Pattern



Observability



Production Architecture Roadmap

